



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS

Course Code & Title: AD3351 Design and Analysis of Algorithms

Name of the Faculty member: Ms.K.Usharani, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Floyd's algorithm
- **Course Outcome:** CO 3 – Implement and analyze the problems using dynamic programming and greedy algorithmic techniques.
- **Topic Learning Outcome:** Discuss the Floyd's algorithm using dynamic programming.
- **Activity Chosen:** Reflection
- **Justification:**
 - The prime objective of Reflection is to make the students write their doubts in a paper on topics which they are not clear, so as the teacher can discuss and clarify the concepts that are not understood by the students.
 - The reflection activity follows the discussion on a particular subject and allows students to evaluate their own understanding of the concept. Students cannot apply floyd's algorithm without a good understanding of the topic.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 1. The students were asked to take a paper and pen.
 2. They were asked to solve the given problems individually based on the previous lecture.
 3. The students were asked to write the doubts and submit the paper.
 4. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.
 5. The instructor evaluates each student's written work to determine their level of understanding of the subject.
 6. On the next day in class, the teacher responds to the student's comments/doubts.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO12	PSO2
CO5	3	1	1	1	2	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO12	PSO2
	3	1	1	1	2	1
Justification for correlation	Apply the knowledge of engineering fundamentals	Identify and analyze the floyd's algorithm in complex engineering problems.	Design solutions for complex engineering problems by understanding the Floyd Warshall's algorithm	Function effectively as an individual by reflecting the ideas on the given topic.	Recognize the need and have the preparation and ability to engage in independent learning.	Motivates Continuous learning to provide the solution to real world business problems.

• **Images / Screenshot of the practice:**

Floyd's Warshall algorithm V. Kousalya

1. difference b/w Floyd's & Warshall
2. How to change from 1-0 By seeing matrix

Floyd's Warshall Algorithm Kousalya, V.

1. Which is All pair shortest path.
2. How to Draw Graph with given matrix in Floyd's.

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students felt that they can able to recall the concepts. And this activity helped them to analyze their level of understanding in the classroom. And also they are able to ask their doubts directly and it get resolved instantly.

❖ **Benefit of the practice:**

1. Students were actively participated in this activity.
2. It increases students understanding on the topic and also improves the student's analyzing ability on the topic.

❖ **Challenges faced in implementation:**

Some students felt difficult to recall the topics on their own.

References:

1. <https://www.clemson.edu/otei/documents/Reflection%20Activities%20r.pdf>
2. <https://teaching.utk.edu/wpcontent/uploads/sites/78/2018/04/ReflectionActivities.pdf>
3. <https://sites.google.com/site/reflection4learning/why-reflect>